MANTEJ SOKHI

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EDUCATION

M.Sc. in Applied Mathematics, Simon Fraser University — Expected 2027
B.Sc. Honours in Applications of Mathematics, University of British Columbia — 2018 – 2023

EXPERIENCE

Graduate Researcher — Sept 2024 - Present
Center for Experimental and Constructive Mathematics — Burnaby, BC

- Designed and built C++ and C libraries for fast finite field polynomial and integer arithmetic and integrated them into Maple through its foreign function interface.
- Implemented rational reconstruction and Newton interpolation in C++, beating Maple's built-ins (Ratrecon and Interp) by **209**× with a 32 bit prime and reducing black box probes by **96**×.
- Implemented Maple's modular Linear solve routine in C++ with batched FFI calls, reducing per call latency from **1075** micro seconds to **6.9** micro seconds (**900**×). Extended the solver to handle 64 bit primes, unsupported by Maple, which is limited to 32 bit hardware integers.
- Designed heuristic and deterministic fault-tolerant variants of the reconstruction algorithms, developing the supporting subresultant theory and failure probability analysis to verify correctness over Galois fields.
- Benchmarked the pipeline on a family of parametric linear systems, outperforming Maple on every case and co-authored a peer reviewed paper (ICMS 2026) on the implementation.

Software Engineer — Feb 2024 - May 2024
Inverted AI — Vancouver, BC

- Built Python cleaning and transformation pipelines that standardized large autonomous driving datasets (100 videos of ∼12 min per country) into consistent, simulation ready inputs for model training.
- Diagnosed recurring CVAT labeling defects by tracing error patterns to their source data and correcting the affected records, stopping **45**% of corrupt annotations from propagating into simulation and training.
- Built Python tools that overlaid annotations onto JOSM base maps, letting the team visually review their drawings and catch geometry errors before validation.

Research Assistant — March 2023 - December 2023
University of British Columbia — Vancouver, BC

- Built Python simulations with NetworkX, NumPy and Pandas to test discrete pursuit evasion strategies across combinatorial graph families, checking that computed damage numbers stayed within derived theoretical bounds.
- Developed and proved the underlying theory and asymptotic bounds, contributing results to a peer reviewed publication.

PROJECTS

Vector Search Engine. Built exact kNN and HNSW search in C++ with Python bindings, multithreaded queries, and index persistence. Achieved **98.2% Recall@10 at 1,850 QPS** on SIFT1M, an **8.4× speedup** over exact search.

Distributed Key-Value Store. Built a concurrent Go TCP store with write-ahead logging, crash recovery, LSM storage, Bloom filters and Raft replication. Sustained **105K ops/s** under a 70/30 workload and **19K replicated writes/s at 14 ms p99** across three nodes with zero acknowledged write loss.

Navier-Stokes Solver. Implemented a C++ 2D lid driven cavity solver on a staggered MAC grid with upwind advection and Jacobi/SOR pressure Poisson solvers. Added residual, divergence and CPU cycle diagnostics.

Polynomial GCD Verification. Formally verified finite field polynomial arithmetic and Euclidean GCD in Dafny with preconditions, loop invariants, termination metrics and proof lemmas. Compiled and tested the verified code across multiple target languages.

TECHNICAL SKILLS

Languages — C++, C, Golang, Maple, Matlab, R, Python, Julia, SQL.
Tools & Infrastructure — Git, Bash, Linux, CMake, Docker, CI/CD Pipelines, Observability (Logging, Metrics, Tracing).